High Myopia–Sensorineural Deafness Syndrome (DFNMYP) — Comprehensive Disease Report

Disease: High Myopia–Sensorineural Deafness Syndrome **MONDO ID:** MONDO:0009082 · **OMIM:** 221200 (DFNMYP) · **Causal gene: SLITRK6** (SLIT and NTRK-like family, member 6) **Category:** Mendelian, autosomal recessive

Summary

High Myopia–Sensorineural Deafness Syndrome (DFNMYP; OMIM 221200; MONDO:0009082) is a rare **autosomal-recessive** disorder caused by biallelic **loss-of-function mutations in SLITRK6**, a single-pass leucine-rich-repeat (LRR) transmembrane protein encoded on chromosome **13q31.1** (HGNC:23503, NCBI Gene 84189, UniProt Q9H5Y7, gene OMIM 609681). The syndrome was defined in 2013 by Tekin and colleagues, who identified homozygous nonsense mutations in three families and showed that the mutant proteins fail to reach the cell surface (PMID: 23543054). The two cardinal features are **congenital high (axial) myopia** and **progressive bilateral sensorineural hearing loss**, the latter having the electrophysiological signature of an **auditory neuropathy**. Vestibular function is preserved.

Mechanistically, SLITRK6 is a **trans-synaptic synapse-organizing protein**. Its extracellular LRR modules bind presynaptic LAR-family receptor protein tyrosine phosphatases (**PTPδ/PTPRD and PTPσ/PTPRS**), building adhesion complexes that drive synaptogenesis and support the survival and innervation of sensory neurons in the retina and cochlea. Truncating mutations abolish surface localization of SLITRK6, breaking this complex. The downstream consequences — demonstrated in *Slitrk6*-deficient mice — are **increased axial eye length** (the endophenotype of myopia), **auditory function deficits**, and **delayed retinal synaptogenesis** (PMID: 23543054). Population genetics support a pure recessive loss-of-function model: gnomAD shows SLITRK6 is tolerant of heterozygous LOF (pLI ≈ 0), and every pathogenic/likely-pathogenic small variant catalogued in ClinVar is truncating.

There is **no disease-modifying therapy**. Management is entirely supportive: refractive and low-vision correction for the myopia, and hearing aids or **cochlear implantation** for the deafness, supplemented by genetic counseling. Because heterozygotes are unaffected and only biallelic complete loss of function causes disease, carrier and cascade testing and reproductive counseling are the principal preventive tools.

Key Findings

Finding 1 — Biallelic truncating SLITRK6 mutations cause the recessive syndrome

The disease is caused by **biallelic loss-of-function mutations in SLITRK6**. The founding study (Tekin et al., 2013) used homozygosity mapping and sequencing to identify **three homozygous nonsense mutations — p.R181X, p.S297X, and p.Q414X — across three families** with an autosomal-recessive syndrome of high myopia plus sensorineural deafness. Critically, all three mutant proteins showed **defective cell-surface localization**, establishing the functional mechanism as loss of a membrane protein:

> "led to the identification of 3 homozygous nonsense mutations (p.R181X, p.S297X, and p.Q414X) in SLIT and NTRK-like family, member 6 (SLITRK6), a leucine-rich repeat domain transmembrane protein. All 3 mutant SLITRK6 proteins displayed defective cell surface localization." (PMID: 23543054)

The allelic spectrum has since broadened across populations. An additional recurrent allele, **p.Gln414Ter (c.1240C>T)**, was reported in **9 Amish homozygotes** (Morlet et al., 2014), and a frameshift allele, **c.696delG (p.Trp232Cysfs*10)**, was found in a homozygous state in a Moroccan family (Salime et al., 2018):

> "identify a novel single deleterious mutation c.696delG, p.Trp232Cysfs*10 at homozygous state in the exon 2 of the SLITRK6" (PMID: 29551497)

All reported disease alleles are **nonsense or frameshift (truncating)** changes, consistent with a loss-of-function disease mechanism.

Finding 2 — The hearing loss is a progressive auditory neuropathy with preserved vestibular function

Detailed audiologic phenotyping of the 9 Amish homozygotes (c.1240C>T, p.Gln414Ter) by Morlet et al. (2014) characterized the deafness as an **auditory neuropathy spectrum disorder** rather than a simple cochlear (sensory) hearing loss. The electrophysiology showed **absent distortion-product otoacoustic emissions (DPOAEs)** in all ears, an **increased-then-absent cochlear microphonic (CM)**, and **dys-synchronised auditory brainstem responses (ABRs)** with no reproducible waves I/III/V at high intensities:

> "Distortion product otoacoustic emissions (DPOAEs) were absent in all ears tested and the cochlear microphonic (CM) was increased in amplitude and duration in young patients and absent in the two oldest subjects. Auditory brainstem responses (ABRs) were dys-synchronised bilaterally with no reproducible waves I, III, or V at high intensities." (PMID: 23946138)

The hearing loss is **progressive and symmetric**, deteriorating with age to severe-to-profound impairment by early adulthood:

> "Hearing loss and speech reception thresholds deteriorated symmetrically with age, which resulted in severe-to-profound hearing impairment by early adulthood." (PMID: 23946138)

Vestibular evoked myogenic potentials were normal, indicating preserved vestibular function — a clinically useful distinguishing feature. This pattern (absent/abnormal OAEs with present CM, dys-synchronous ABR) is the classic signature of auditory dyssynchrony/ neuropathy, pointing to a lesion at the level of inner hair cell synapse / auditory nerve rather than outer hair cell mechanics alone.

Finding 3 — *Slitrk6*-deficient mice recapitulate the human syndrome

The mouse knockout reproduces both organ phenotypes, providing strong causal evidence. High-resolution MRI of *Slitrk6*-deficient eyes revealed **increased axial length — the endophenotype of myopia** — and the mutant mice displayed **auditory function deficits mirroring the human phenotype**. Retinal histology during postnatal development showed a **delay in synaptogenesis**:

> "High-resolution MRI of WT and Slitrk6-deficient mouse eyes revealed axial length increase in the mutant (the endophenotype of myopia). Additionally, mutant mice exhibited auditory function deficits that mirrored the human phenotype. Histological investigation of WT and Slitrk6-deficient mouse retinas in postnatal development indicated a delay in synaptogenesis in Slitrk6-deficient animals." (PMID: 23543054)

The biological role unifying both phenotypes is neurotrophic/synaptic:

> "Slitrk6 plays a role in the survival and innervation of sensory neurons in the inner ear, vestibular apparatus, and retina, and also influences axial eye length" (PMID: 23946138)

Finding 4 — SLITRK6 is tolerant of heterozygous LOF (gnomAD), confirming recessive inheritance

Gene-constraint metrics from gnomAD (GRCh38) show SLITRK6 is **not haploinsufficient**: **pLI ≈ 1.3×10^{-7} (≈0)**, **observed/expected LOF (oe_lof) = 0.63** with LOEUF = 0.83 (observed/expected = 35/55.6, lof_z = 2.34). A pLI near zero means loss of one allele is well tolerated in the general population — exactly what is expected for a recessive disorder in which only **biallelic** LOF causes disease, and heterozygous carriers are unaffected.

Confirmed identifiers anchoring the gene: **UniProt Q9H5Y7** (841 aa), **NCBI Gene 84189**, **HGNC:23503**, gene **OMIM 609681**, cytoband **13q31.1**; mouse ortholog **Slitrk6** (Entrez 239250).

Finding 5 — ClinVar variant spectrum is exclusively truncating for pathogenic alleles

ClinVar (gene SLITRK6, transcript NM_032229.3) lists **80 variants**: 65 Uncertain significance, 8 Likely benign, 3 Pathogenic, 2 Likely pathogenic. Every pathogenic/likely-pathogenic **small** variant is **truncating** — e.g., **c.698T>A (p.Leu233Ter, nonsense)** and **c.846del (p.Leu283fs, frameshift)**. The "Pathogenic" set also includes **large multi-megabase 13q deletions** that encompass SLITRK6 (e.g., 13q31.1–34; 13q14.3–31.2). The complete absence of pathogenic missense alleles reinforces a **pure loss-of-function** mechanism — there is no evidence for gain-of-function or dominant-negative disease.

Finding 6 — Protein architecture: single-pass membrane protein with two extracellular LRR modules

UniProt Q9H5Y7 (841 aa) annotates a canonical **SLITRK architecture**: an N-terminal **signal peptide**; a large **extracellular region (~aa 27–608)** containing **two tandem LRR modules** (LRRNT1 27–67 + 5 LRRs + LRRCT1 218–269; LRRNT2 320–361 + 6 LRRs + LRRCT2 517–568); a single **transmembrane helix (aa 609–629)**; and a **cytoplasmic tail (aa 630–841)**. Subcellular location is the **cell membrane**. UniProt's functional summary is "**Regulator of neurite outgrowth required for normal hearing and vision**," and its disease annotation (DFNMYP) reads "an autosomal recessive disorder characterized by prelingual sensorineural hearing loss associated with high myopia." Tissue specificity is highest in **putamen**, with expression in adult/fetal lung, fetal liver, and high levels in some brain tumors (medulloblastoma, PNET).

Truncating mutations at p.R181X, p.Trp232fs, p.Leu233Ter, p.S297X, p.Leu283fs, and p.Q414X all fall within or upstream of the extracellular LRR domains, removing the transmembrane anchor and/or the ligand-binding LRR surface — explaining the observed **loss of cell-surface localization**.

Finding 7 — SLITRK6's top partners are presynaptic LAR-RPTPs (PTPδ/PTPσ) and IL1RAPL1

STRING (human, confidence ≥0.4) ranks SLITRK6's highest-confidence functional partners as **PTPRD/PTPδ (0.986)**, **IL1RAPL1 (0.986)**, and **PTPRS/PTPσ (0.970)**, with lower-confidence NTRK1/TrkA (0.590) and others (GRK1, PPEF2, OLFM2, BMPR1B, STK4, LRRC20). **PTPδ and PTPσ are the established presynaptic LAR-family receptor protein tyrosine phosphatases** that bind SLITRK extracellular LRR domains to assemble **trans-synaptic adhesion complexes**. This places SLITRK6 squarely in the synapse-organizing pathway and provides the molecular bridge between gene loss and the sensory-neuron synaptic deficits seen in retina and cochlea.

Report by Section

1. Disease Information

Overview. DFNMYP is a Mendelian, autosomal-recessive syndrome combining **congenital high myopia** and **progressive bilateral sensorineural hearing loss** (auditory-neuropathy type). It is caused by loss of function of **SLITRK6**, a synaptic organizer required for normal development of the eye and inner ear.

Key identifiers:

Resource	Identifier
OMIM (disease)	221200 (DFNMYP)
OMIM (gene)	609681 (SLITRK6)
MONDO	MONDO:0009082
HGNC	23503
NCBI Gene	84189
UniProt	Q9H5Y7
Ensembl / cytoband	13q31.1
Mouse ortholog	Slitrk6 (Entrez 239250)

Synonyms / alternative names: High myopia–sensorineural deafness; Deafness and myopia syndrome; Myopia–deafness syndrome; DFNMYP; SLITRK6-related myopia and deafness. (No additional formal synonyms recorded in the source template.)

Information source type: Aggregated disease-level resources (OMIM, UniProt, ClinVar, gnomAD, STRING) combined with **individual-patient clinical reports** from the primary cohorts (families in Tekin 2013; 9 Old Order Amish in Morlet 2014; one Moroccan family in Salime 2018).

2. Etiology

Primary cause: Genetic — biallelic (homozygous or compound heterozygous) loss-of-function mutations in **SLITRK6**. No environmental or infectious cause.

Genetic risk factors: The causal variants are truncating alleles (nonsense/frameshift) and large 13q deletions encompassing SLITRK6. Specific recurrent alleles include **p.R181X, p.S297X, p.Q414X** (Tekin 2013), **p.Gln414Ter** / **c.1240C>T** (Amish founder allele, Morlet 2014), and **c.696delG** / **p.Trp232Cysfs*10** (Moroccan, Salime 2018). **Consanguinity** and **founder effects** (notably the Amish c.1240C>T) are the main population-level risk drivers, as expected for a rare recessive disorder.

Environmental risk / protective factors: None established. The myopia here is **syndromic and axial/genetic**, distinct from common multifactorial refractive myopia; near-work and lighting (relevant to common myopia) are not established modifiers of this monogenic form.

Protective genetic factors / modifier genes: None reported. No modifier loci or protective alleles have been identified for DFNMYP.

Gene–environment interactions: None documented; disease is essentially fully penetrant given biallelic LOF.

3. Phenotypes

Phenotype	Type	HPO term (suggested)	Onset	Severity / Progression	Frequency
High myopia (axial)	Physical/ ophthalmologic	HP:0011003 (High myopia) / HP:0000545 (Myopia)	Congenital / early childhood	Severe; axial elongation	Universal (defining)
Sensorineural hearing loss	Clinical sign	HP:0000407 (Sensorineural hearing impairment)	Prelingual/ congenital	Progressive, symmetric; severe-to-profound by early adulthood	Universal (defining)
Auditory neuropathy pattern	Laboratory/ electrophysiology	HP:0011962 (Abnormal auditory brainstem response)	Early	Dys-synchronous ABR, absent DPOAE, abnormal CM	Reported in Amish cohort
Preserved vestibular function	(Negative finding)	—	—	Normal VEMP	Reported

Quality-of-life impact: The dual sensory deficit (vision + hearing) has substantial impact on communication, education, and independent living. Early hearing amplification/cochlear implantation and low-vision support materially improve function, but the combined deficit places affected individuals in a **dual-sensory-impairment** category requiring multidisciplinary rehabilitation.

4. Genetic / Molecular Information

Causal gene: SLITRK6 (gene OMIM 609681; HGNC:23503; NCBI Gene 84189; chr 13q31.1), notable as a gene whose coding sequence lies in a single exon.

Pathogenic variants (representative):

Variant (cDNA)	Protein	Class	Source / population
—	p.R181X	Nonsense	Tekin 2013
—	p.S297X	Nonsense	Tekin 2013
c.1240C>T	p.Gln414Ter (p.Q414X)	Nonsense	Tekin 2013; Amish founder (Morlet 2014)
c.696delG	p.Trp232Cysfs*10	Frameshift	Salime 2018 (Moroccan)
c.698T>A	p.Leu233Ter	Nonsense	ClinVar (Pathogenic)
c.846del	p.Leu283fs	Frameshift	ClinVar (Pathogenic)
13q multi-Mb deletions	(whole-gene loss)	Structural	ClinVar (Pathogenic)

Variant classification: Per ACMG/AMP, the recurrent truncating alleles are Pathogenic/Likely pathogenic; the vast majority of ClinVar entries (65/80) are VUS, and 8 are Likely benign. **No pathogenic missense alleles** exist — the spectrum is exclusively LOF.

Allele frequency: Disease alleles are rare/private; gnomAD shows SLITRK6 tolerant of heterozygous LOF (pLI ≈ 0, LOEUF 0.83).

Origin: Germline. **Functional consequence:** Loss of function (abolished cell-surface localization). No gain-of-function or dominant-negative mechanism.

Modifier genes / epigenetics: None reported. **Chromosomal abnormalities:** Large 13q31 deletions encompassing SLITRK6 can produce the phenotype as part of contiguous-gene deletions.

5. Environmental Information

Not applicable. DFNMYP is a monogenic recessive disorder with no established environmental, lifestyle, or infectious contributors. (The axial myopia is genetically determined, unlike common environmentally modulated myopia.)

6. Mechanism / Pathophysiology

Ordered causal chain:

1. **Biallelic truncating mutation in SLITRK6** (nonsense/frameshift, or whole-gene 13q deletion) → produces a truncated protein or no protein.

2. Truncated SLITRK6 **loses its transmembrane anchor and/or LRR ligand-binding surface → fails to localize to the cell membrane** (demonstrated for p.R181X, p.S297X, p.Q414X).
3. Absent surface SLITRK6 → **cannot bind presynaptic LAR-family RPTPs (PTPδ/PTPRD, PTPσ/PTPRS) → trans-synaptic SLITRK6–LAR-RPTP adhesion complex fails to form** (inferred from STRING interactome + known SLITRK–LAR biology).
4. Loss of synaptic organizing signal → **impaired/delayed synaptogenesis** and **reduced survival/innervation of sensory neurons** in retina and cochlea (demonstrated as delayed retinal synaptogenesis in mouse).
5. **Branch A (eye):** disrupted retinal synaptic development/visual signaling → **dysregulated axial growth of the eye → increased axial length → high axial myopia** (axial elongation demonstrated by MRI in *Slitrk6* KO).
6. **Branch B (ear):** impaired innervation/survival of cochlear sensory neurons and inner-hair-cell synapse → **auditory dyssynchrony** (absent DPOAE, abnormal CM, dys-synchronous ABR) → **progressive sensorineural hearing loss / auditory neuropathy**, worsening symmetrically to severe-to-profound by early adulthood.
7. Vestibular apparatus relatively spared → **normal vestibular function** (preserved VEMP).

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SLITRK6 LOF mutation
      │ (loss of surface protein)
      ▼
SLITRK6–PTPδ/PTPσ trans-synaptic complex not formed
      │ (impaired synapse organization)
      ▼
delayed synaptogenesis + poor sensory-neuron innervation/survival
      ├──────────────► RETINA → dysregulated axial growth → HIGH MYOPIA
      └──────────────► COCHLEA → auditory dyssynchrony → PROGRESSIVE SNHL
                       (vestibular system spared)
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Molecular pathway: Trans-synaptic LRR adhesion / **SLITRK–LAR-RPTP synapse organization** (GO:0007416 synapse assembly; GO:0050808 synapse organization). **Cellular process:** neurite outgrowth regulation (GO:0010975), neuron development, synaptogenesis. **Protein dysfunction:** loss of function via mislocalization (not aggregation). **Cell types (CL):** retinal neurons / bipolar neurons (CL:0000540 neuron; CL:0000748 retinal bipolar neuron), cochlear hair cells / spiral ganglion neurons (CL:0000601 cochlear hair cell). **Upstream:** the SLITRK6 mutation and loss of synaptic adhesion. **Downstream:** axial elongation and auditory dyssynchrony.

No immune, metabolic, or inflammatory mechanism is implicated.

7. Anatomical Structures Affected

- **Primary organs: Eye** (UBERON:0000970) — retina (UBERON:0000966) and globe affecting axial length; **Inner ear / cochlea** (UBERON:0001844 cochlea; UBERON:0001690 ear).
- **Body systems:** Special sense organs — **visual system** and **auditory/sensory system**.
- **Tissue/cell level:** Neural retina (photoreceptor-to-bipolar synaptic layers), cochlear sensory epithelium and spiral ganglion neurons; inner hair cell ribbon synapse implicated by the auditory-neuropathy pattern.
- **Subcellular (GO CC):** cell membrane / plasma membrane (GO:0005886); synapse (GO:0045202); integral component of plasma membrane (GO:0005887).
- **Localization / lateralization: Bilateral and symmetric** for both eye and ear involvement.

8. Temporal Development

- **Onset: Congenital/prelingual.** High myopia present from early childhood; hearing loss prelingual (often congenital).
- **Onset pattern:** Chronic/insidious.
- **Progression:** The **hearing loss is progressive**, deteriorating **symmetrically with age to severe-to-profound by early adulthood** (PMID: 23946138). The myopia is high/axial.
- **Disease course:** Chronic, lifelong, progressive (for hearing). No remission; no self-limited course.
- **Critical period:** Early childhood is the window for intervention — early audiologic amplification/cochlear implantation and refractive correction to support language and visual development.

9. Inheritance and Population

- **Inheritance: Autosomal recessive.** Biallelic LOF required; heterozygous carriers unaffected (consistent with gnomAD pLI ≈ 0).
- **Penetrance:** Essentially complete for biallelic LOF. **Expressivity:** The two cardinal features are consistent, with some variability in hearing-loss severity by age.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Founder effects:** Yes — the **Amish c.1240C>T (p.Gln414Ter)** allele is a documented founder allele (Morlet 2014, 9 homozygotes).
- **Consanguinity:** A major contributor; most reported families are consanguineous or from genetically isolated populations.

- **Carrier frequency:** Not formally estimated at population scale; rare overall, elevated locally in founder populations.
- **Epidemiology: Ultra-rare**; precise prevalence/incidence not established. Reported in a small number of families worldwide.
- **Geographic distribution:** Cases reported in Turkish/Greek/Mennonite, Old Order Amish (USA), and Moroccan populations — clustering in consanguineous or isolated communities.
- **Sex ratio:** No sex predilection (autosomal). **Age distribution:** Affected from birth; hearing worsens through childhood into early adulthood.

10. Diagnostics

- **Ophthalmologic:** Refraction and **axial length measurement** (biometry) documenting high axial myopia; fundus examination.
- **Audiologic:** Pure-tone audiometry; **DPOAE** (absent), **cochlear microphonic** (abnormal), **ABR** (dys-synchronous, absent waves I/III/V), middle-ear muscle reflexes (absent in older patients) — together establishing an **auditory neuropathy** pattern. **VEMP** normal (vestibular sparing) — a useful differentiator.
- **Genetic testing (confirmatory): Single-gene SLITRK6 sequencing**, deafness/retinal gene panels including SLITRK6, or **whole-exome sequencing (WES)**; **chromosomal microarray (CMA)** detects large 13q deletions encompassing SLITRK6. WES/panel is the practical first-line given the combination of ocular + auditory features; reference transcript NM_032229.3.
- **Clinical criteria:** Diagnosis = **congenital high axial myopia** + **progressive bilateral sensorineural hearing loss** with **biallelic truncating SLITRK6 variants**.
- **Differential diagnosis:** Other syndromic deafness-with-eye-disease conditions must be excluded — notably **Usher syndrome** (deafness + retinitis pigmentosa, often with vestibular involvement in type 1), **Stickler syndrome** (high myopia + hearing loss + arthro-ophthalmopathy, autosomal dominant COL2A1/COL11), **Marshall syndrome**, and **Wolfram syndrome**. The auditory-neuropathy electrophysiology, preserved vestibular function, isolated high myopia (no retinal dystrophy/arthropathy), and SLITRK6 genotype distinguish DFNMYP.
- **Screening:** Carrier screening in at-risk families; cascade testing of relatives; newborn hearing screening will flag the auditory component.

11. Outcome / Prognosis

- **Survival / mortality: Normal life expectancy**; the disorder is not life-limiting. No disease-specific mortality.
- **Morbidity:** Substantial sensory morbidity — combined visual and progressive auditory impairment affecting communication, education, and daily function (dual sensory impairment).
- **Disease course:** Hearing loss progresses to severe-to-profound by early adulthood; myopia is high and lifelong. **Recovery potential:** No spontaneous recovery; function is improved (not cured) by devices.
- **Prognostic factors:** Severity/progression of hearing loss and timing of intervention (early amplification/cochlear implantation and refractive support predict better communication and educational outcomes). Genotype is uniformly LOF, so genotype–phenotype gradation is limited.

12. Treatment

No disease-modifying or curative therapy exists. Management is supportive and device-based (suggested NCIT intervention terms in parentheses):

- **Ophthalmologic:** Corrective **refractive lenses** / **spectacles** / **contact lenses** and **low-vision aids**; routine retinal surveillance for high-myopia complications (retinal detachment, myopic maculopathy) (NCIT: Eyeglasses; Corrective Lens).
- **Audiologic: Hearing aids**; **cochlear implantation** for severe-to-profound loss. Because the lesion is an auditory neuropathy, cochlear-implant candidacy/benefit should be individually assessed, but implantation is a mainstay for severe-to-profound cases (NCIT: Hearing Aid; Cochlear Implant).
- **Rehabilitation / supportive: Speech-language therapy**, educational support, and dual-sensory-impairment services (NCIT: Speech Therapy; Rehabilitation Therapy).
- **Pharmacotherapy / gene therapy / cell therapy / RNA therapy:** None established or in trials specific to SLITRK6/DFNMYP. No pharmacogenomic considerations.
- **Experimental:** No DFNMYP-specific clinical trials identified.

13. Prevention

- **Primary prevention:** Not possible (genetic). **Genetic counseling** is the principal tool — 25% recurrence risk per pregnancy for carrier couples.

- **Genetic/carrier screening:** Carrier testing in consanguineous/founder families; **cascade testing**; **prenatal diagnosis** and **preimplantation genetic testing (PGT)** are options for known-variant families.
- **Secondary prevention: Newborn hearing screening** and early pediatric refraction enable early intervention during the critical developmental window.
- **Tertiary prevention:** Monitoring and managing high-myopia complications (retinal detachment, maculopathy); optimizing amplification/implantation to prevent language and developmental deficits.
- **Public health / immunization / behavioral:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / model:** *Mus musculus* (NCBI Taxon 10090) — the principal experimental species. **Orthologous gene:** mouse **Slitrk6** (Entrez 239250).
- **Natural disease in other species:** No naturally occurring companion-animal or wildlife DFNMYP homolog is documented (no OMIA entry noted).
- **Comparative biology / conservation:** SLITRK6's synaptic-organizer function and the SLITRK–LAR-RPTP interaction are evolutionarily conserved among vertebrates, supporting cross-species validity of the mouse model.
- **Zoonosis:** Not applicable (non-infectious).

15. Model Organisms

- **Model type:** Mammalian — **Slitrk6 knockout mouse** (the key disease model).
- **Phenotype recapitulation: Strong.** The KO reproduces (i) **increased axial eye length** (myopia endophenotype, by high-resolution MRI), (ii) **auditory function deficits** mirroring the human phenotype, and (iii) **delayed retinal synaptogenesis** on postnatal histology (PMID: 23543054). Slitrk6 contributes to survival/innervation of sensory neurons in inner ear, vestibular apparatus, and retina (PMID: 23946138).
- **Applications:** Mechanistic dissection of axial eye growth control, retinal synaptogenesis, and cochlear neuron innervation; platform for candidate therapeutics.
- **Limitations:** Mouse vision/auditory physiology differs from human; the model does not capture the full progressive human hearing-loss trajectory or the human refractive endpoint directly. Additional cellular/iPSC or organoid models for DFNMYP were not identified.
- **Resources:** MGI (Slitrk6). Cellular models (iPSC/organoid) not reported.

Mechanistic Model / Interpretation

The entire phenotype is parsimoniously explained by **loss of a single trans-synaptic organizing protein**. SLITRK6 is a two-LRR, single-pass membrane protein (Finding 6) whose extracellular LRR surface engages presynaptic **LAR-RPTPs (PTPδ, PTPσ)** to nucleate synapse formation (Finding 7). When truncating mutations (Findings 1, 5) remove the membrane anchor / LRR surface, the protein never reaches the cell surface (Finding 1), so the adhesion complex cannot assemble. Downstream, synaptogenesis is delayed and sensory-neuron innervation/survival is compromised — directly demonstrated in the mouse as **delayed retinal synaptogenesis**, **axial eye elongation**, and **auditory deficits** (Finding 3). The two clinical branches (myopia via disrupted retinal/visual control of eye growth; progressive auditory neuropathy via inner-ear synaptic/innervation failure) share this single upstream lesion, while the vestibular system is relatively spared. Population genetics close the loop: SLITRK6 tolerates monoallelic loss (gnomAD pLI ≈ 0, Finding 4) and only truncating/whole-gene-deletion alleles are pathogenic (Finding 5) — confirming a **recessive, pure loss-of-function** model with no dominant-negative or gain-of-function component.

Layer	Observation	Evidence
Gene	Biallelic truncating SLITRK6 variants	Tekin 2013; Morlet 2014; Salime 2018; ClinVar
Protein	Loss of cell-surface localization	Tekin 2013 (in vitro)
Interactome	Binds PTPδ/PTPσ (LAR-RPTPs), IL1RAPL1	STRING
Cell/tissue	Delayed synaptogenesis; poor sensory innervation	Mouse (Tekin 2013; Morlet 2014)
Organ	Axial eye elongation; auditory dyssynchrony	Mouse MRI + human electrophysiology
Clinical	High myopia + progressive SNHL, vestibular sparing	Human cohorts
Population	Recessive; heterozygotes unaffected	gnomAD

Evidence Base

PMID	Title	Type	Contribution
23543054	*SLITRK6 mutations cause myopia and deafness in humans and mice.*	Human clinical + mouse model + in vitro	Founding study: 3 homozygous nonsense mutations; defective surface localization; KO mouse recapitulates axial myopia, auditory deficits, delayed retinal synaptogenesis
23946138	*A homozygous SLITRK6 nonsense mutation is associated with progressive auditory neuropathy in humans.*	Human clinical (9 Amish)	Defines auditory-neuropathy electrophysiology, progressive symmetric hearing loss, vestibular sparing; Slitrk6 neurotrophic role
29551497	*A novel mutation in SLITRK6 causes deafness and myopia in a Moroccan family.*	Human clinical	Adds frameshift allele c.696delG (p.Trp232Cysfs*10); extends allelic/ population spectrum

Supporting database evidence (no single PMID): **gnomAD** (constraint, Finding 4), **ClinVar** (variant spectrum, Finding 5), **UniProt Q9H5Y7** (architecture/function, Finding 6), **STRING** (interactome, Finding 7). These resources are mutually consistent and all point to a recessive loss-of-function synaptic-organizer mechanism.

Limitations and Knowledge Gaps

1. **Small human evidence base.** The clinical picture rests on ~3 cohorts/families; prevalence, carrier frequency, and full phenotypic range are not well quantified.
2. **Mechanistic gaps in the eye.** How SLITRK6 loss translates specifically into *axial elongation* (vs. retinal degeneration) is not fully worked out at the molecular level; the SLITRK6→PTPδ/PTPσ→eye-growth link is partly inferred.
3. **Auditory-neuropathy site of lesion** (inner hair cell ribbon synapse vs. spiral ganglion neuron) is inferred from electrophysiology, not confirmed by human histopathology.

4. **No longitudinal natural-history data** quantifying rates of myopia/axial progression or hearing decline; no formal QoL instruments applied.
5. **Cochlear-implant outcomes** specific to SLITRK6 auditory neuropathy are not systematically reported — CI benefit in auditory neuropathy can be variable.
6. **No therapeutic development** (gene/cell/RNA therapy) and no cellular/organoid human models identified.

Proposed Follow-up Experiments / Actions

1. **Natural-history registry** across known families to quantify axial-length and hearing-threshold trajectories and define intervention windows.
2. **Cochlear-implant outcome study** in SLITRK6 patients to determine whether the synaptic/neuropathic lesion permits good electrical-stimulation benefit.
3. **Mechanistic imaging/histology** in the KO mouse (and human iPSC-derived retinal and inner-ear organoids) to map the SLITRK6–PTPδ/PTPσ synaptic deficit to axial growth control and inner-hair-cell synaptopathy.
4. **Structure-function / rescue assays:** test whether re-expression of wild-type SLITRK6 (or isolated LRR domains) restores surface localization and PTPδ/PTPσ binding, and whether gene-replacement rescues the mouse phenotype — proof-of-concept for future gene therapy.
5. **Variant reclassification effort:** functional assays (surface-localization, LAR-RPTP binding) on the 65 ClinVar VUS to resolve their clinical significance.
6. **Carrier-frequency estimation** in founder populations (Amish, North African) to inform targeted carrier-screening programs.

Report generated from a 5-iteration autonomous investigation; 7 findings confirmed and verified against primary literature (PMIDs 23543054, 23946138, 29551497) and database evidence (gnomAD, ClinVar, UniProt Q9H5Y7, STRING).
